



Group Projects Structure and Systems Bioinformatics

- Systems Bioinformatics Section –

Project 1

Benchmark of comparative single-cell analysis approaches

Tutor:

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In the lecture we discussed comparative single-cell analysis tasks such as the detection of disease associated cell subsets. Specifically, we discussed two approaches to achieve this task. The first approach consisted of deriving cell types by unsupervised learning (clustering) and training a classifier over frequencies of the resulting cell types. The second approach constitutes an end-to-end solution based on convolutional neural networks (CellCnn).

In this group project the goal is to (1) revisit the original publication introducing the latter approach and (2) reimplementing it and the two reported baseline approaches and (3) assessing and documenting classification results by means of dimensionality reduction techniques.

For the execution of the project we will provide you with the original publication

Arvaniti E, Claassen M. Sensitive detection of rare disease-associated cell subsets via representation learning. Nature communications. 2017 Apr 6;8(1):1-0.

and a single-cell proteomic dataset of immune cells that has been recorded for a patient collective reflecting two clinical conditions. Please perform the following tasks

1. **Literature research:** Familiarize with the paper. Summarize the approaches for comparative single-cell analysis presented and benchmarked in the paper.
2. **Dimensionality reduction:** Download the provided dataset. Visualize the dataset with PCA/t-SNE/UMAP. What are the parameters that you can choose for each dimensionality reduction technique? What is the impact of these parameters on the visualization. Document example visualizations. What quantitative measures could you use to compare the difference of the visualizations? Choose one measure and assess it for all pairs of visualizations.
3. **Clustering:** Cell type identification typically resorts to clustering approaches. Assess the clustering results for three different clustering approaches, i.e. k-means, hierarchical clustering and Leiden. Which of the clustering solutions does best reflect biological cell types? What quantitative criterion could you use to guide this decision. Consider reading up the original publication reporting the dataset.
4. **Comparative analysis benchmark:** Implement the three comparative single-cell analysis approaches presented in the paper. Assess classification performance for each approach.
5. **Comparative analysis interpretation:** Define a quantitative score identifying the cell subset determining the result of the classifiers. Use your dimensionality reduction approach of choice to visualize these cell subsets.
6. **Bonus task - Improving comparative analysis:** How could you modify the CellCnn model to achieve better classification performance? For instance consider variations of the filter response function.

In your final report, please summarize everything you did for your project. Your final report should be 5 pages long including figures, tables, and references.